

Augmenting Indigenous Sámi Exhibition - Interactive Digital Heritage in Museum Context

Siiri Paananen, Joo Chan Kim, Emma Kirjavainen, Matilda Kalving, Karan Mitra, Jonna Häkkilä

▶ **To cite this version:**

Siiri Paananen, Joo Chan Kim, Emma Kirjavainen, Matilda Kalving, Karan Mitra, et al.. Augmenting Indigenous Sámi Exhibition - Interactive Digital Heritage in Museum Context. 19th IFIP Conference on Human-Computer Interaction (INTERACT), Aug 2023, York, United Kingdom. pp.597-617, ⟨10.1007/978-3-031-42283-6_32⟩. ⟨hal-05757120⟩

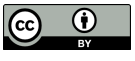
HAL Id: hal-05757120

https://ifip.hal.science/hal-05757120v1

Submitted on 21 Sep 2026

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons CC BY 4.0 - Attribution - International License



This document is the original author manuscript of a paper submitted to an IFIP conference proceedings or other IFIP publication by Springer Nature. As such, there may be some differences in the official published version of the paper. Such differences, if any, are usually due to reformatting during preparation for publication or minor corrections made by the author(s) during final proofreading of the publication manuscript.

Augmenting Indigenous Sámi Exhibition - Interactive Digital Heritage in Museum Context

Siiri Paananen[1(✉)], Joo Chan Kim[2], Emma Kirjavainen[1], Matilda Kalving[1], Karan Mitra[2], and Jonna Häkkilä[1]

[1] University of Lapland, Rovaniemi 96300, Finland
{siiri.paananen,emma.kirjavainen,matilda.kalving,jonna.hakkila}@ulapland.fi
[2] Luleå University of Technology, Skellefteå 93177, Sweden
{joo.chan.kim,karan.mitra}@ltu.se

Abstract. Museums and cultural heritage institutions have an important role in presenting accurate information and sharing cultural knowledge, and new technologies are increasingly implemented. For the best results, the appropriateness of a specific technology must be evaluated for each context. Research has shown the need for participatory methods and local knowledge in Indigenous design contexts. We describe a case study where an Indigenous Sámi museum exhibition was augmented with interactive technology through multidisciplinary co-design work with museum experts, designers, and developers. The traditional clothing of the Sámi people was digitized by filming, and information related to it was presented as a touchscreen installation in a renewed exhibition. User tests including interactive tasks and interviews (n = 7) and a questionnaire (n = 27) were completed on-site. The installation was rated interesting and easy to use, while some users struggled to find all the features. Our study shows that a technically relatively simple digital installation can be easy-to-use but interesting for a standard museum visitor. Additionally, the work demonstrates how to build successful collaborations that highlight Indigenous cultural heritage. We discuss the implications of using technology to promote cultural heritage and identities.

Keywords: Interactive systems · User studies · Digital cultural heritage · Museum exhibitions · Indigenous HCI

1 Introduction

Museums have an essential role when societies' histories and identities are considered. They provide information about the past but also reflect what are con-

The work has been supported by Interreg Nord project 'Muittut, muitalusat - the story of the Sámi by the Sámi' and 'Innovation in Lapland through Design and Art' project, funded by the European Regional Development Fund (ERDF). We wish to thank Ájtte, the Swedish Mountain and Sámi Museum. Special thanks to Sunna Kuoljok and Ann-Catrin Blind.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2023
J. Abdelnour Nocera et al. (Eds.): INTERACT 2023, LNCS 14143, pp. 597–617, 2023.
https://doi.org/10.1007/978-3-031-42283-6_32

sidered the building blocks of society, community, or nation and their culture. The role has both external and internal functions [21], as the museum exhibitions project an image of the community to the outsiders visiting the place, but are also part of the community's identity building. Thus, how museum exhibitions reflect the identity and culture of a group or an entity is essential. Museums are limited by their size and resources. Hence a variety of solutions can be considered, depending on the case. Technologies can assist museums in creating engaging and educational interactive content [35], e.g. with multi-sensory applications. The designers and developers of the technologies in a cultural context have to pay special attention to make the process fair and beneficial to the museum.

This paper describes a case study of developing a digital touchscreen program for an Indigenous Sámi museum, presenting traditional clothes. The motivation was to showcase Sámi cultural heritage using digital tools and to provide equal opportunities for sharing their histories and stories in an interactive and accessible way in the museum. The study acts as an example of a practical case study, where a part of the material heritage of Indigenous Sámi culture was applied as a digital museum exhibit. The program was co-designed iteratively with the museum and two universities working together, and aimed to showcase the clothes and their history and meaning by renewing an existing exhibition. The Sámi perspective of the museum experts was there to guide the design process from the first concepts to the final product. While there have been technologies applied to Sámi museums, not much has been published from the point of view of HCI researchers based on our research. Sámi museums face a challenge in representing the culture and the aspects of modernity/traditionality [48].

In this study, our target was to evaluate how a new program and the renewed exhibition were received by museum visitors. We were also interested in how the participants perceived combining the new technology and cultural heritage and how the participants felt about the image of the culture projected by the exhibition. In this paper, we also describe the issues that were part of the development process and how those were solved, thus sharing the lessons learned. Our work allows the HCI community to get insight into the design context of Indigenous museums and consider taking into account different cultural settings and applying technologies in the wild in a culturally sensitive design context. Our work contributes to providing different cultures with a better chance of equal opportunities and allows technology to cater for fruitful cross-cultural exchanges.

The paper is structured as follows; first, related work to HCI development and evaluation in Indigenous and museum contexts is presented, after which the design process and method are described. Then the results are presented, followed by a discussion, including lessons learned and notes on how the program was improved, then finally going through the study's limitations and conclusions.

2 Related Work

Our work takes place in the cross-section of design and development as well as Indigenous cultural and museum contexts. Historical events and political currents affect people's views and feelings on cultural heritage and museums. When

designing technologies for Indigenous communities and museums, the design context, ethics, and cultural sensitivities need to be considered [11]. One perspective which can be applied is decolonizing design, see e.g. Tlostanova [45]. Parsons et al. [36] have studied the collaborative aspects of Indigenous and academic work, and suggested a pluralistic way of looking at Indigenous and Western worldviews. Lazem et al. [26] write, "Indigenous knowledge is an important resource, and we should utilise the tools and methods that we have at our disposal to enhance it." When highlighting the need for ethical processes, Lazem et al. also call for researchers to apply flexibility, respect, and sensitivity towards the local culture. A literature review we previously conducted on working with technology and cultural heritage through decolonizing design highlights the need for participatory methods and working with local values and cultural contexts [34].

2.1 Indigenous Context and HCI

The Indigenous Sámi collections were transferred from the Museum of Ethnography to *Ájtte, the Swedish Mountain and Sámi Museum* in Jokkmokk, when it was opened in 1989 as the only Sámi museum in Sweden. Museums and their work are an important part of identity building; as Silvén [43] writes, "During more than hundred years Sámi representation in heritage and museums has contributed to defining Sámi identity and the position of Sámi in society". Silvén has also later described that while it is positive that Sámi people are recognized and made more visible, e.g. in museums, new interpretations should also be made so that the image of Sámi is not locked only into the history [44]. In 2008 Harlin [12] suggested that for Sámi Cultural Heritage, collecting information on the existing objects and their location can be a step in the process of repatriation, which means returning cultural heritage back to the community. Lately, there have been efforts to gather information about the collections of Sámi Heritage around Europe and bring them under one search portal service for better accessibility [10]. This has been made possible by the digitalization of archives and new digital tools, allowing users to see old pictures of traditional objects and clothes. Traditions and their presentation have been researched e.g. by Cocq [7], who focuses on the Sámi in Sweden. They write about traditionalization and its relationship to revitalization, exploring how traditions are presented online by different institutions and their effect on people's identities [7]. Aikio [1] has looked at photographs in the old exhibition of Sámi Museum Siida and analyzed how *Sáminess* was presented there, noting that the old exhibition did not include enough of contemporary Sámi life, but shows it as something in the past. The museum renewed its exhibitions in 2022 and involved Sámi communities in the process [1].

Indigenous HCI as a concept has been around for a while, [18] but appears to be somewhat still lacking in the broader HCI community discussion. A workshop organized by Lawrence et al. [25] was aimed at anyone involved or interested in the context, and included a definition; "Indigenous HCI involves work across all aspects of the computing ecosystem including education, research, design, development and implementation." Different approaches can be applied when

working with Indigenous communities and HCI topics. For instance, cultural adaptation in user interface (UI) design has been demonstrated and applied by integrating local objects and components into the GUI metaphor design [14,52]. Co-design and Indigenous-led processes have been suggested. Magnani et al. [30] have presented a community-based method of making digital 3D models of Sámi cultural heritage objects. An approach to HCI design projects with Indigenous people leading them has been described by Peters et al. [38], who do not think mere participation is enough. Their approach includes user-leadership in the co-design process led by Indigenous university students. One suggested way of exploring and promoting Sámi identities and intangible cultural heritage by the community has been game development, e.g. through the event Sámi Game Jam by Laiti et al. [24]. Their approach involves Sámi people participating in game-making and telling stories through interactive digital media: video games. Heimgärtner [13] has suggested a model for cultural influence on HCI, including the local and Indigenous contexts. Different cultures have varying perspectives on concepts such as time, communication, and space, which should be considered in the design and development process. Through co-creation methods, Indigenous knowledge can be conveyed into digital technologies [49]. Zaman and Falak [53], who have sought to create an Indigenous HCI paradigm, emphasize the need for designers to engage with the local community. While technology can be adapted to different contexts, in some cases, applying digital technologies has been entirely discarded in the design process due to them not fitting the goal of the projects or the local community goals, as noted by Kelly and Taffe [22] in their case study with Indigenous community museum. Their involvement with co-design work led to the understanding that the digital solutions would not be optimal in the presented scenario, but it is noted that they should be evaluated case-by-case. Thus, technology should not be leading the design process.

2.2 Designing and Evaluating Digital Heritage Applications

A plethora of studies exists that have addressed the different aspects of designing interactive technologies for museums and exhibitions and evaluating their usability and UX. The area has been of interest to many researchers, and designing for cultural heritage fosters an active research community in HCI [2,6]. Much of the prior art has presented examples and case studies of integrating digital content and interactive technologies into museums and reported the advantages of interactive exhibitions. For instance, Hornecker and Stifter [17] have addressed interaction design in museum exhibitions. They note how the interactive installations with tangible physical elements interested all kinds of users. Our earlier study on museum experiences confirmed the need for interactivity and noted the importance of considering ethical issues [33]. Konstantakis and Caridakis [23] suggest that cultural heritage institutions should consider the User Experience before, during, and after the visit for better satisfaction and impact on visitors. They conducted a survey to represent different aspects related to Cultural User Experience by looking at 37 applications in the field. They also note the importance of the user's expectations, which will affect their actual perception of the experience.

In our research, we especially focus on presenting digital 3D content as part of the museum exhibition. A case study by Micoli et al. [31] shares the design process of an interactive touchscreen application with 3D content and hotspots, explaining the different content levels and relations between the selected objects and the different levels of the program. The content for their app is a complex museum collection, and they suggest giving the visitors connections between the physical exhibition and digital information. When comparing the visitor's experience between attending physical and virtual exhibits, Wolf et al. [50] report the pros and cons. Whereas VR was appreciated for allowing access from anywhere, anytime, the VR experience was criticized by participants for losing the authentic atmosphere.

The prior art has also addressed the methodological approaches to studying museum experiences. Hornecker and Nicol [16] have performed a comparison of user studies both in a lab setting and in the wild with the same museum game application on touchscreens. They note how the context of the study matters and details such as the installation setup or group behaviours and interactions. Konstantakis and Caridakis [23] also note how instead of only lab studies, field studies are needed to understand the audience. Karoulis et al. [20] conducted a survey and a walkthrough session to evaluate a system with experts. They suggest that domain experts (e.g. museum curators) and usability experts should be utilized when conducting expert usability evaluations for museum applications for best results. Barbieri et al. [3] have proposed a user study methodology for virtual museums, in their case, comparing two design alternatives for interactive 3D content presented on a screen. Their methodology included metric tasks, a questionnaire as well as an interview. Using multiple methods allows for understanding the results more deeply.

3 Design Context and Process

In this section, we share the background of the design context, as well as the design processes.

Sámi culture. The Indigenous Sámi people live in Sápmi, in the Northern parts of Norway, Sweden, Finland, and Russia, with many subgroups and languages (e.g. Valkonen et al., [46]). The research and design process presented in the following focuses on the theme of the Sámi dress, Gapta (Gákti in Northern Sámi). The dress is traditional Indigenous clothing but continues evolving with the culture, like Magga [29, p.50] writes, 'A living gákti tradition implies a living Sámi culture. And vice versa.' Thus, the Gapta is a central part of the living Sámi culture and identity building and in a key role when a museum exhibition is considered. The exhibition included nine showcases, each with 2-3 Gaptas from different regions of Sápmi. This also meant that the designed program was presented on nine different screens next to the showcases. The aim of the exhibition was both to show the old dresses, as well as modern skills in dressmaking and

evolving traditions, showing that the culture is still living (Fig. 1). During the development process, three seminars were organized for understanding the Sámi culture for the outsiders of the Sámi community.

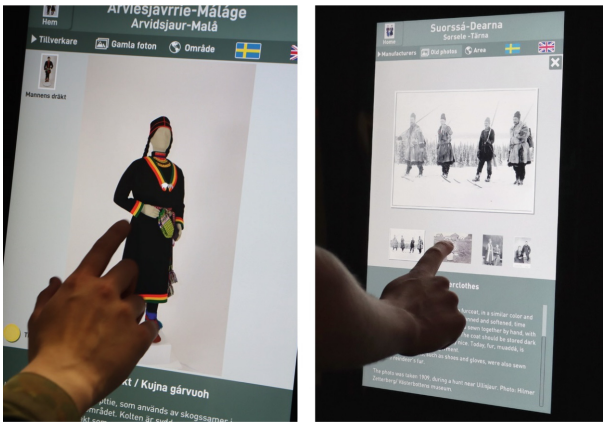


Fig. 1. Testing the program's UI in the lab (left) and at the museum (right).

The Development Process. We organized three teams, each with different roles in the program development (Fig. 2). First, the programming team developed the technical part of the program with Unity 2022. Second, the design team worked on the graphics and UI, including the program's UI layout, icon design, typography, usability, and user flow. Third, the museum experts evaluated whether the program fulfilled the requirements for their exhibition, provided the content for the program, and finally acted as quality assurance testers, identified any errors to fix, and confirmed the program's quality. Initially, various technologies were proposed (e.g., holography, augmented reality, and virtual reality); however, the ideas using them were discarded due to a disaccord between the technologies, the context of the exhibition, and the selected topic. Collaborating with Sámi culture experts was educational and it helped the designers to understand cultural factors that needed to be considered in addition to usability and technology, for a successful outcome. Special care had to be put into ethical perspectives, in order to support the Sámi experts' visions and highlight their culture in an inclusive way. Trust and good communication were essential parts of the collaboration in order to learn and understand the cultural meanings and their significance. A similar iterative five-step framework for developing a game in a museum context has been described by Gestwicki and McNely [9], and the authors highlight that regular meetings with the project team encourage empathy.

During the development, each team co-operated to improve the program's quality iteratively. For example, after the initial concept was decided, the programming team worked with the design team to concretize the program layout

and user flow. Meanwhile, the design team communicated with the quality assurance team to clarify the design requirements. The quality assurance team then talked with the programming team to confirm whether a desired feature for the program was feasible within resources. A weekly meeting was held with all three groups to synchronize the development process, share updates on each team's status, and identify more improvements for our program. The work was conducted online because of the COVID-pandemic and long distances, but the team members were able to meet during the final user testing and data gathering.

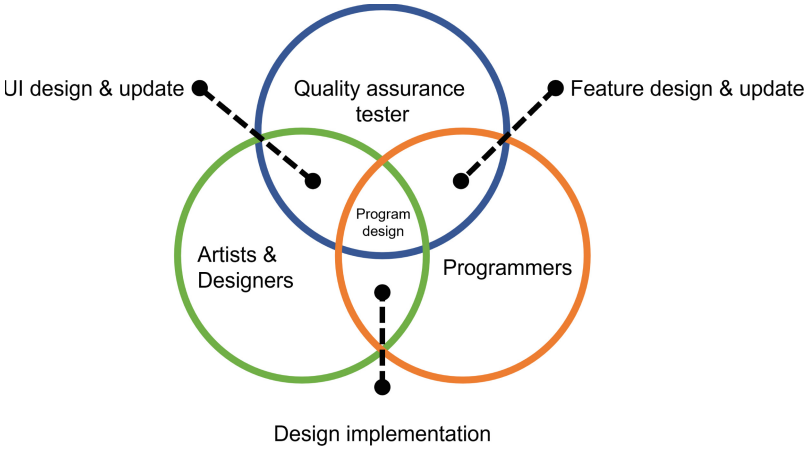


Fig. 2. Roles for each team in the program development.

The dresses are quite detailed and have a lot of different textures and patterns. Instead of using 3D models, which require professional skills (e.g., modelling and texturing) to create with a certain level of quality for the exhibition, the dresses were filmed as 360-degree videos. The resulting video was then implemented in Unity to appear as a 360-object, which can be observed from all sides by rotating with the user's touch gesture. Considering the environmental context and readability of Gapta content, we used nine Windows PCs with 17-inch touch screens with 1920×1080 resolution to run our program for the Gapta exhibition. The hardware was hidden behind the walls and showcases to avoid hindrance to user engagement for the exhibition.

User Experience and User Interface. In the UI design process, much attention was paid to the wishes of the museum experts, as they know the content best. The UI was designed to suit the museum's style regarding fonts and colours. The exhibition room was renovated, therefore, the new program's visual look was considered to fit the space and aesthetic. In addition to the visual look, issues related to usability had to be considered, such as how readable texts and pictures are. Special attention was paid to ensuring that the program was inclusive

and usable from visual and cognitive standpoints and physical aspects, such as the user's height. The prototype was tested on a simulated colourblindness site and several font sizes to consider different users. The design focused on assisting people in navigating the program by using icons and animations, for example, on the hotspots, encouraging users to notice them. Special care was put into the graphic design of the map, which shows the area where the clothes are traditionally used. As Sápmi hasn't traditionally regarded country borders, the designed map only featured a dotted line as the borders between Finland, Sweden, and Norway.

We used a virtual prototyping method (e.g. Wang [47]) during the design process. It helped to focus on what the UI must enable in interaction scenarios. Virtual prototyping created an opportunity to test and develop the UI iteratively and allowed the museum experts to test the prototypes remotely. We decided to utilize only touch-based interaction due to the environmental context. Firstly, Remote gesture-based interaction (e.g., hand or full body tracking) were not applied in order to avoid too challenging a learning phase for interaction. Touch-based interaction is a familiar modality for people in the smartphone era; hence, we employed it as the primary interaction modality. Secondly, since the nine programs work in the same space, audio-based interaction (i.e. voice command and auditory output) would cause a jarring scene with overlapping sounds. Since hearing is one of the five human senses, sounds could take a huge role in an overall user experience either positively or else [19,27,51].



Fig. 3. Visitor using the touchscreen in the museum exhibition, with the physical Gaptas in a showcase.

4 Method

Here, the methodologies applied to the conducted user studies are described. We conducted two types of user evaluation to gain a better understanding of how the visitors received the renewed exhibition and program. The first was a

questionnaire comprising general questions about the museum exhibition and the short version of the User Experience Questionnaire (UEQ) [42]. Another evaluation was individual interviews and complimentary surveys (SUS) [5] with exhibition visitors. The two evaluations had a different set of participants from each other.

Questionnaire. We prepared a three-page questionnaire, including a consent form to ask for participants' agreement for data collection. The questionnaire was left for two weeks near the Gapta exhibition (Fig. 3) with an advertisement for the museum visitors to fill it in. The participants were asked to return their answered questionnaire to the museum reception before they left. The questionnaire had 27 participants, ages 16 to 79, with a a mean of 41,8. Most of the participants identified as female (78%), 19% as male, and one participant as "other". Six participants had never visited the museum exhibition before, while 13 participants had only visited the old exhibition, one only the renewed exhibition, and seven participants had visited the old and the new one. 19 people reported that they were from Sweden and the rest from Europe and Africa.

The first part of the questionnaire included background questions and general questions related to the exhibition and its subjective usability. The next page featured UEQ, and to reduce the answering time for the museum visitors, we used the short version of UEQ, which only has 8 out of the 26 items from the original version. Four items ("obstructive/supportive," "complicated/easy," "inefficient/efficient," and "confusing/clear") in the short version UEQ represent a pragmatic quality related to perceived usability. The other four items ("boring/exciting," "not interesting/interesting," "conventional/inventive," and "usual/leading edge") represent a hedonic quality of user experience. Participants rated their impression of the program on a 7-step scale from −3 to 3. Selecting a negative value meant the participant thought the negative term fit the experience. In contrast, a positive value referred a positive term. When the participant felt none of the terms fit, they were asked to select zero as neutral. The questionnaire also included additional questions to identify helpful design elements for using the program. In addition, we asked whether the museum visitors would have wanted to use a different interaction modality for input (e.g., a voice command, or body gesture) and output (e.g., music or speech).

We utilized the analysis tool provided by UEQ developers [15]. With this tool, we could get the means of each item, quality, and overall evaluation. The results were also benchmarked based on the data from 21,175 persons from 468 studies to check how well the program performed [15]. The additional answers were then coded [41] to find which elements were helpful for using the program and what is the museum visitors' opinion regarding the interaction modality of the program.



Fig. 4. The interview data was organized by affinity mapping

Interview and Observation. For deeper qualitative insights we conducted one-on-one interview-based usability tests with seven participants. The participants were recruited from museum visitors by both the researchers and the museum staff. The interviews were conducted in a private room at the museum, where the participants were presented with a screen that resembled the one used in the exhibition. All the participants (n=7) were female, with six Swedish and one Finnish participant. One of the participants reported their age to be between 26-35, one participant was 36-45, another 56-65, and the rest, four participants were 66+ years old. When it comes to the demographics, it is worth noting that 215 people visited the museum during the two days of user testing, of which 190 were locals. The local town's population of around 4,800 is mainly seniors over 65 years old. One of the participants had never visited the museum before, while four had already visited the old exhibition and two the new one. One usability test session lasted approximately 20 min, and the participants were compensated for their time with a small gift from the museum shop.

The in-depth user study process consisted of the following stages:

- 1) Initial briefing about the study and filling out the consent form,
- 2) performing three tasks and participating in a semi-structured interview,
- 3) filling out the end questionnaire and System Usability Scale (SUS)

Informed consent was explained by the researcher and signed by the participant before the study was conducted. The participant was tasked with finding the manufacturer of the clothing, a map, and a close-up photo of the lower part of the jacket or shoes. Their interactions with the touchscreen were recorded and counted. Immediately after the participants finished the tasks, we conducted a semi-structured interview about their experience, the touchscreens' perceived usability, and functions. The interview was audio-recorded and transcribed. It was followed by a short questionnaire, where the participant filled out their background information and answered three open-ended questions. Finally, the

participant filled out the System Usability Scale (SUS), which is used to measure subjective usability. It consists of a ten-item questionnaire with a 5-point Likert scale, from strongly agree to strongly disagree. The SUS score was counted, and the interviews were analysed by affinity mapping [28,32] the answers and observations (Fig. 4). The users' performance was calculated by the number of interactions with the touchscreen. The SUS score and performance results were validated by comparing them to their individual performance.

5 Results

Here, the results of the two data sets are shared, first the general questionnaire and then the user testing session, including an interview, observation, and survey.

5.1 Questionnaire

While there were 27 participants total, one participant stopped answering the questionnaire after the background questions and one general question and wrote that the screens did not work. The questionnaire respondents were presented with three 5-Step Likert-scale questions related to the exhibit;

- The Gapta exhibit was interesting (n = 27). mean = 4,74
- The Gapta exhibit was easy to use (n = 26). mean = 4,73
- I feel like the Gapta exhibit is part of my culture (n = 23). mean = 4,05

A few participants pointed out in the third question that they were not Sámi, thus unable to respond.

When asked about their feelings on combining technology with old museum objects, 23 participants responded positively, such as: *"Good. The technology provides opportunities for extensive information on a small and discreet surface"* (Participant 8) and *"Super! Easy to navigate and immerse themselves in what you think is interesting"* (P18). Three felt neutral or had mixed opinions, saying *"It's good as long as it works"* (P23) and *"I would say it wouldn't hurt to combine technology with the museum object as long as it doesn't override"* (P4). When asked for their first impressions of the exhibition, the participants expressed things related to visuals, such as colours, lights, information, culture, dresses, and atmosphere. When asked for other comments on the topic, participants reported various things, such as general notes on the exhibits, opening hours, or missing information. A few participants expressed pride related to the culture being showcased well; *"I like to see our culture highlighted in such a good way"* (P16), *"Feeling proud of this cultural, historical heritage"* (P7). One participant reported that the touch function was hard to use (P22), and another pointed out related to the dolls and their gender and that it was nice to see the children, not only with women (P20). One person commented that they missed the old texts from the previous exhibition, *"..they showed the important things we do not always talk about in Sápmi"* (P17).

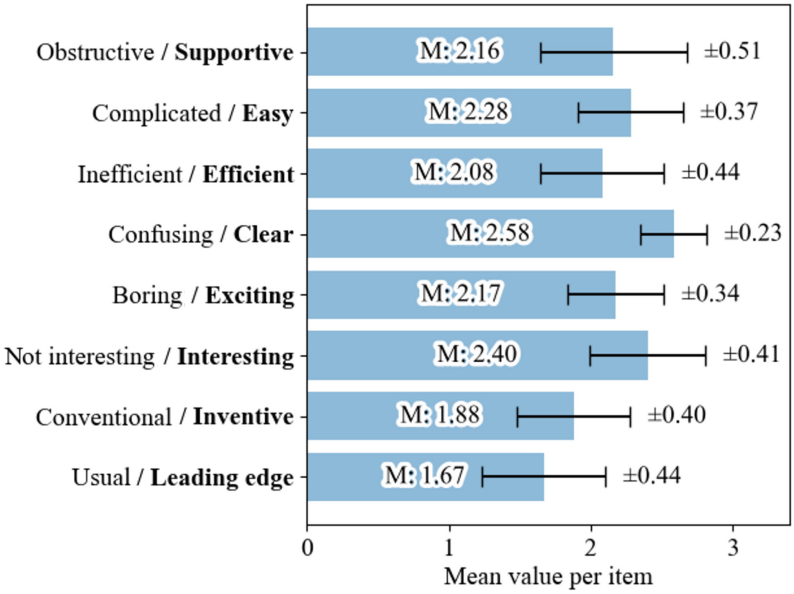


Fig. 5. The mean values of each item in the user experience questionnaire (UEQ).

Regarding the UEQ data, two participants (P11 and P25) were removed from the analysis due to missing answers. One participant (P25) skipped answering the additional questions about interaction modality. In total, 25 participants' questionnaire responses were used for the UEQ data analysis, and 26 participants' responses to the questions about interaction modality were examined. We got a mean of over 1.67 for all items from a scale of −3 to 3. The item "confusing/clear" received the highest mean of 2.58, followed by "not interesting/interesting" with a mean of 2.40. On the other hand, the item "usual/leading edge" was rated with the lowest mean of 1.67. The second lowest item was "conventional/inventive" with a mean of 1.88. Figure 5 depicts the means and 95% confidence interval of each item. Since all the items received positive results, they are highlighted with bold text. The pragmatic quality received 2.27, while the hedonic quality achieved 2.03 on a scale of -3 to 3. Consequently, our program was evaluated positively for both pragmatic and hedonic aspects, with a mean of 2.15 for the overall quality. The benchmark result (Fig. 6) suggested that our program is rated positively for all qualities compared to other programs. The results of every item reached an "Excellent" grade, the highest grade in the benchmark.

For helpful elements chosen by the visitors, icons were found to be the most helpful for navigating the program by 23 participants, colour differences by seven participants and familiar designs by "six" participants. Altogether 25 participants answered yes when asked if they liked controlling the screen with touch, and two (P22 and P25) did not respond. Some of the explanations for enjoying

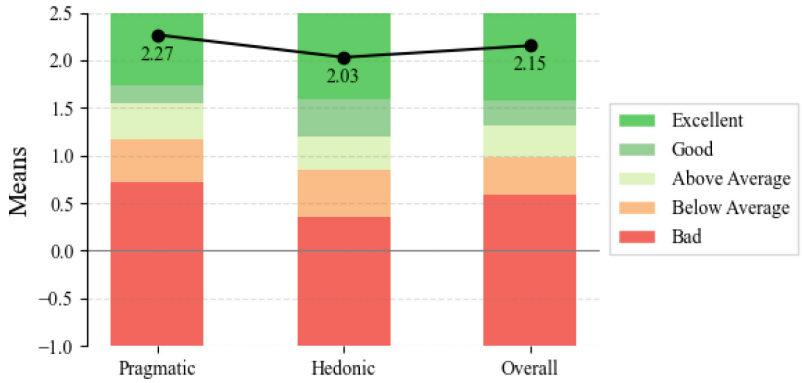


Fig. 6. The benchmark result, based on 21,175 persons' answers from 468 studies.

it included it being easy (P4, P1, P11, and P12), being able to choose what to read or see (P15, P21, P23, and P27), *"To be able to see the 3D model"*, probably meaning the rotatable dress (P20), feeling activated (P24), and participating (P14). The participants were asked whether they would like to use another way to avoid touching the screen, and eight agreed. Four selected hand gestures as an alternative way, two chose body gestures, and one selected "other" and wanted to use a pointer. 21 participants answered "yes" when asked if the texts were easy to read; three gave a mixed answer, one said no, and another did not respond. Regards to the text, the responses indicated that it was clear (P5, P12, P18, and P23), the font size was right (P4, P13, and P22), it had good resolution (P4) but was too long (P2 and P22), spacing was dense (P24) and scrolling to read the text was a negative experience (P1 and P27). 14 participants wished that the pilot would contain audio. Five people wrote they wanted to hear music (P2, P3, P4, P15, and P23), two wished for joik, which is traditional Sámi singing (P5 and P12), speech was requested by three participants (P11 and P14), and also the need for the texts to be read aloud was expressed by two participants (P17 and P18), a narration was wished by two; one who also liked to have music (P4) and another participant wanted an artisan to explain about their process (P9). The last participant, who answered that there was no need for audio, noted that headphones would be needed if there were any sounds (P8).

5.2 Interview, Observation and Survey

As part of the interview analysis, the qualitative information was categorized and sorted using an affinity diagram, called affinity mapping. We noted that the participants' previous technical knowledge affected their use of the touchscreen, as two participants had previous work experience that supported learning new functions and searching for information. In contrast, one participant had no technical knowledge and required guidance to use the screen. The participant could not finish the procedure without help from the facilitator. The different

technical knowledge levels helped us comprehensively understand the museum visitors' needs. Overall the participants scored the touchscreen very highly on the SUS, with an overall score of 85, "excellent" (Table 1). The participant with the lowest rated SUS score (62.5, good) was also the one with the least experience with technology. Even when she noticed the added value of the touchscreen, she would not necessarily use it in her everyday life, as she stated, *"You look if you are interested, as simple as that."* (Interview Participant 2). When analyzing performance, time, observation, and clicks had to be considered. We found that some users progressed by touching the touchscreen immediately, while others read the whole page and interacted only after considering the options.

Table 1. Table of each participant's time and clicks required for completing the tasks, with the SUS score to compare.

Participant		**Manu-factorer**		**Map**		**Lower part of the jacket or shoes**		**SUS**
nr	age	clicks	sec	clicks	sec	clicks	sec	score
IP1	26-35	10	58	1	12	26	40	87.5
IP2	66+	8	158	1	16	4	50	62.5
IP3	36-45	1	7	1	3	10	26	90
IP4	56-65	2	9	1	1	27	92	85
IP5	66+	1	7	1	6	7	145	92.5
IP6	66+	1	3	2	6	4	83	85
IP7	66+	1	5	1	4	3	48	95
	Mean:	**3.43**	**35.29**	**1.14**	**6.86**	**11.57**	**69.14**	**85.36**

With the help of the survey results and affinity mapping, we were able to evaluate the perceived usability and learnability of the touchscreens. Two things stood out very clearly in the results. Firstly, recognizable functions improve the participant's immersion and comprehension of the interface and secondly, unrecognizable icons lead to the participants not finding and using functions that would have made the use easier or opened up new sections in the program, e.g. stating after task 3 (find the lower part of the jacket or shoes), *"it is clear that it[the experience] will be completely different if knowing it[the function]"* (IP7). The problem was mostly remarked on the sliding gesture and hotspots. As a result of the participants not understanding the rotating mechanism, they could not comprehend and use hotspots, which led to them never discovering the close-up photos on their own. During the analysis, we noted and categorized the visual aspects that stood out in the data. Three participants mentioned that the interface was either calm, discreet, or non-intrusive, and most participants mentioned the size or amount of pictures as a positive influence on their experience. The neutral, photo-centric interface was believed to present the images more clearly, as one participant stated, *"The picture was the one that caught my eye. It's probably because of the layout"* (IP1). While the participants appreciated the amount of data included in the application, it also exhausted them.

As for innovations and suggestions for improvements, the participants expressed a need for more languages, such as Sámi languages. At the time of testing, there was only Swedish and English present. Participants deemed English necessary, but adding more languages could add to the accessibility and inclusivity. Some participants had knowledge of the local culture and identity, e.g. one participant recognized a familiar name on the list of manufacturers and was visibly delighted. One of the participants pointed out that the height of the screens should always be set to the standard and be adjustable for better visibility, e.g. stating *" ..I would like to be able to adjust the height where the screen is placed. The screen's viewing angles are such that when the screen is not at eye level, the colours start fading at the edge"* (IP1).

6 Discussion

The design process and the results of user experience evaluation in situ using multiple methods have been presented. Next, we reflect on our findings and design considerations and discuss the limitations. Both UEQ and SUS data showed "excellent" results, suggesting that the program's user experience and perceived usability worked well in the wild. Some features were not easily found by the participants in the interview and user evaluation. The positive aspects reported by the participants were the visual aspects and the amount and size of photos, and the familiar icons and interaction method were seen as helpful. The results also indicate that people could connect to the culture (e.g. feeling proud) when looking at it on display. When designing interactive screens in a museum environment, one should take accessibility, understandability, and learnability into consideration, in addition to the cultural aspects. In the co-design process, listening to the substance experts was found critical when choosing the direction to proceed with development.

6.1 Indigenous Museum as a Design Context

Indigenous museums are a particular design context, requiring special care and methods. The research presented in this paper was part of a larger work effort focusing on developing a shared *Sámi Museum Language*, to be used in the future for all the Sámi museums working on the topic, featuring museum partners from Finland, Sweden, and Norway. The process involved creating shared icons for the museums and imagining and prototyping different ways of presenting cultural heritage content in museums with digital technologies. Perhaps through co-design methods, interactive technologies can provide a contemporary interpretation of the Sámi culture, which Silvén [44] calls for in order to create new meanings for traditions. To enable safe and fruitful collaboration, much thought had to be put into the design work from ethical and cultural viewpoints [11]. This case example operates as a starting point for the work, but further work needs to be done, as there are multiple museums with varying needs, resources and cultural areas. As noted in the literature [23], both usability and content experts are needed to evaluate an interface; therefore, multidisciplinary development should be encouraged.

Some of our results indicated the feeling of cultural identity and the participants being proud of seeing their culture shown positively. Thus, museums can have a remarkable role when building a picture of peoples, cultures, and areas, both for the community and outsiders. As emphasized by Harlin [12], for the Sámi people, material culture, including objects such as clothes, is important and a way to express things like ethnicity and where one is from. So they can be an important way of expressing identity, which can also be showcased through digital storytelling and enhancing traditional physical exhibits with technology. The literature suggested that sometimes digital technologies can be left out entirely [22] if a suitable way of including them is not found.

6.2 Familiarity Supporting the User

We found that the overall user experience in terms of pragmatic and hedonic aspects is positive for the program. Two items in the pragmatic quality, "confusing/clear" and "complicated/easy," received the highest positive values (i.e., 2.58 for "clear" and 2.28 for "easy"), and we identified potential factors that affect the user experience from additional questions. The icon was selected by 88% of participants (23 out of 26 participants), which could imply that the icon significantly influenced the positive results. A few participants chose the colour differences for UI components and familiar UI layout designs. Since familiar design elements (e.g., UI layout and icon design) are helpful in building an intuitive interface [4,39], the icon design and UI layout are designed to present a friendly image that the participants might have experienced before. The questionnaire results implied that the UI designed with familiar images might help participants get used to the program through an easy experience. The participants preferred to interact with their fingers due to the intuitiveness and familiarity of touch interaction. Although some participants sought an alternative interaction modality instead of touching a screen due to the concern of hygiene after the COVID-19 pandemic, more than half of the participants (17 out of 26) preferred to use touch interaction. Hand or body gesture-based interactions require knowing specific motions to give a command properly.

6.3 Typography for Multi-line Text

We identified that font size and text alignment are impactful elements for reading in a dim exhibition space. As bigger font size and larger line spacing could improve the readability [40], tests were performed on readability starting with size 20 font, 1 line spacing, and various design elements in typography, including font size, colours, and line and letter spacing. A virtual prototype was used to test different sizes and types to find a font size that is readable even if the user wears glasses or their eyesight is not good.

The screens are installed at a height of about 100 cm to enable interaction for people of different heights, therefore, the taller participants have to look downward. We concluded that the font size 30 with 1 line spacing and sans serif typeface is suitable for texts with multiple lines. White colour for the text and dark

green for the background was used to give a higher contrast. Positive feedback was received regarding the font size for reading from most participants. However, some noted low readability for various reasons, including lengthy text and typography-related issues. People are diverse in terms of height, sight, concentration, and preferences. Accordingly, a possible solution to improve the readability could be to support customizable font sizes with different line spacing [8].

6.4 Updates and Design Considerations for the Future

After the user tests were conducted at the exhibition, the UI and the system received small changes, taking into consideration the results (Fig. 7). The biggest changes included having the top buttons look more like buttons to increase the UI's intuitiveness and slowing down the rotation speed for easier interaction and a better view of the Gapta's details and adding a 360-icon under the doll to indicate that it can be rotated freely with a swipe gesture. Small adjustments were also made, such as an updated icon and changing the term for manufacturer to artisan for clarity. The participants wished for the inclusion of Sámi language(s), which was the initial plan but was cut out for resource reasons. We hope that in the future, they can be added. Other desired features included adding audio and narration, which the museum could consider.

Although the program aimed to offer inclusive design [37] for visitors of all skill levels, a participant not being able to finish the usability testing tasks without assistance could indicate that touch-based interaction method (i.e., press and swipe) might be too much for some. Perhaps they can benefit from a shared experience and join others who navigate the technology in a shared experience. Several participants wished for audio output while viewing the content in the program. We decided to omit any sounds from our program due to the environment in the exhibition room. However, if a directional speaker could be used, the program could enrich the overall experience Furthermore, adding another channel to deliver the content to the museum visitor could support people with disabilities, or if a supported interaction modality is not applicable to the museum visitor.

6.5 Limitations

We acknowledge that our work is limited by the number of participants. In our study, the demographic was not balanced in genders and ages, but the regular visitors of the museum have to be considered. As this was a study only on one exhibition, more extensive work could be conducted on different exhibits and museums to get a broader view of the topic. Also, as mentioned earlier, users' expectations of the experience will affect their perception of the experience [23], and as our participants had already chosen to go to the museum, they might have some positive expectations on the visit.

None of the authors in this paper was Indigenous; thus, we approached the topic as outsiders looking in. We communicated actively regarding the program design between the UI designers, program developers, and Sámi museum experts. An even better understanding of each other's domain could have been included if we had more chances to share knowledge.

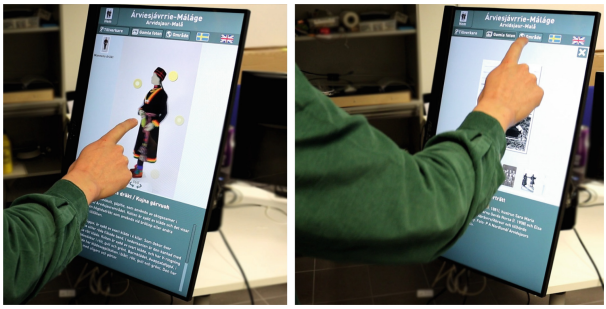


Fig. 7. The program being tested after making the changes.

7 Conclusion

Our case study has shown an example of an HCI development process in the Indigenous design context. Our results implied that the program was received well by the participants in general as part of the renewed exhibition. We found some perceived usability issues related to features and finding them, which were then used to update the program. We also presented suggestions on what to develop in the future, if possible. We discussed the meaning of applying technologies to promote cultural identities and heritage, allowing people to learn, appreciate and recall cultural aspects at a museum. There needs to be more work between different disciplines in order to showcase Indigenous cultures in museums in the best way possible.

The process was based on the needs of the museum and the Sámi community, focusing on clear information and simple interactions. We feel this topic is essential in making the HCI community think about specific design considerations related to Indigenous HCI topics, e.g. when working with Indigenous museums. We hope we can give them insight, on what things to focus on in such a context. More work is needed around the world in regard to Indigenous HCI.

References

1. Aikio, Á.: A window into vanishing sámi culture?: visual representations of sáminess in the shared siida exhibition by sámi museum siida and northern lapland's nature centre 1. In: The Sámi World, pp. 21–38. Routledge (2022)
2. Antoniou, A., De Carolis, B., Kuflik, T., Origlia, A., Raptis, G., Gena, C.: AVI-CH 2022: workshop on advanced visual interfaces and interactions in cultural heritage. In: Proceedings of the 2022 International Conference on Advanced Visual Interfaces, pp. 1–3 (2022)
3. Barbieri, L., Bruno, F., Muzzupappa, M.: Virtual museum system evaluation through user studies. J. Cult. Herit. **26**, 101–108 (2017)
4. Blackler, A., Popovic, V., Mahar, D.: Intuitive interaction applied to interface design. In: New Design Paradigms: Proceedings of International Design Congress (IDC) 2005, pp. 1–10. International Design Congress (2005)

5. Brooke, J., et al.: SUS-a quick and dirty usability scale. Usability Eval. Indus. **189**(194), 4–7 (1996)
6. Ciolfi, L., Damala, A., Hornecker, E., Lechner, M., Maye, L., Petrelli, D.: Cultural heritage communities: technologies and challenges. In: Proceedings of the 7th International Conference on Communities and Technologies, pp. 149–152 (2015)
7. Cocq, C., et al.: Traditionalisation for revitalisation: tradition as a concept and practice in contemporary sámi contexts. Folklore: Electr. J. Folklore **57**, 79–100 (2014)
8. Darroch, I., Goodman, J., Brewster, S., Gray, P.: The effect of age and font size on reading text on handheld computers. In: Costabile, M.F., Paternò, F. (eds.) INTERACT 2005. LNCS, vol. 3585, pp. 253–266. Springer, Heidelberg (2005). https://doi.org/10.1007/11555261_23
9. Gestwicki, P., McNely, B.: A case study of a five-step design thinking process in educational museum game design. In: Proceedings of Meaningful Play (2012)
10. Häkkilä, J., Paananen, S., Suoheimo, M., Mäkikalli, M.: Pluriverse perspectives in designing for a cultural heritage context in the digital age. In: Artistic Cartography and Design Explorations Towards the Pluriverse, pp. 134–143. Routledge (2022)
11. Häkkilä, J., et al.: Design sensibilities-designing for cultural sensitivity. In: Proceedings of the 11th Nordic Conference On Human-computer Interaction: Shaping Experiences, Shaping Society, pp. 1–3 (2020)
12. Harlin, E.K.: Repatriation as knowledge sharing-returning the sámi cultural heritage. Utimut: Past Heritage-future Partnerships: Discussions On Repatriation in the 21st Century, pp. 192–200 (2008)
13. Heimgärtner, R.: Reflections on a model of culturally influenced human-computer interaction to cover cultural contexts in HCI design. Int. J. Hum.-Comput. Inter. **29**(4), 205–219 (2013)
14. Heukelman, D., Obono, S.E.: Exploring the African village metaphor for computer user interface icons. In: Proceedings of the 2009 Annual Research Conference of the South African Institute of Computer Scientists and Information Technologists, pp. 132–140 (2009)
15. Hinderks, A., Schrepp, M., Thomaschewski, J.: User Experience Questionnaire (UEQ) (2018). https://www.ueq-online.org/
16. Hornecker, E., Nicol, E.: What do lab-based user studies tell us about in-the-wild behavior? insights from a study of museum interactives. In: Proceedings of the Designing Interactive Systems Conference, pp. 358–367 (2012)
17. Hornecker, E., Stifter, M.: Learning from interactive museum installations about interaction design for public settings. In: Proceedings of the 18th Australia Conference on Computer-human Interaction: Design: Activities, Artefacts and Environments, pp. 135–142 (2006)
18. Hugo, J., Day, B.: Business: government, business, HCI, and technological development in South Africa. Interactions **8**(4), 25–32 (2001)
19. Hutchinson, R., Eardley, A.:'I felt i was right there with them': the impact of sound-enriched audio description on experiencing and remembering artworks, for blind and sighted museum audiences. Museum Management and Curatorship, pp. 1–18 (2023)
20. Karoulis, A., Sylaiou, S., White, M.: Usability evaluation of a virtual museum interface. Informatica **17**(3), 363–380 (2006)
21. Kelly, L.: Measuring the impact of museums on their communities: the role of the 21st century museum. Intercom **2**(4), 11 (2006)
22. Kelly, M., Taffe, S.: When digital doesn't work: experiences of co-designing an indigenous community museum. Multimod. Technol. Inter. **6**(5), 34 (2022)

23. Konstantakis, M., Caridakis, G.: Adding culture to UX: UX research methodologies and applications in cultural heritage. J. Comput. Cult. Herit. (JOCCH) **13**(1), 1–17 (2020)
24. Laiti, O., Harrer, S., Uusiautti, S., Kultima, A.: Sustaining intangible heritage through video game storytelling-the case of the Sami game jam. Int. J. Herit. Stud. **27**(3), 296–311 (2021)
25. Lawrence, C., Leong, T.W., Brereton, M., Taylor, J.L., Bidwell, N., Wadley, G.: Indigenous HCL: Workshop at Ozchi 2019, Perth. In: Proceedings of the 31st Australian Conference on Human-Computer-Interaction, pp. 17–19 (2019)
26. Lazem, S., Giglitto, D., Nkwo, M.S., Mthoko, H., Upani, J., Peters, A.: Challenges and paradoxes in decolonising HCI: a critical discussion. Comput. Support. Cooperat. Work (CSCW) **31**, 1–38 (2021). https://doi.org/10.1007/s10606-021-09398-0
27. Loureiro, S.M.C., Roschk, H., Lima, F.: The role of background music in visitors' experience of art exhibitions: music, memory and art appraisal. Int. J. Arts Manag. **22**(1), 4–24 (2019)
28. Lucero, A.: Using affinity diagrams to evaluate interactive prototypes. In: Abascal, J., Barbosa, S., Fetter, M., Gross, T., Palanque, P., Winckler, M. (eds.) INTERACT 2015. LNCS, vol. 9297, pp. 231–248. Springer, Cham (2015). https://doi.org/10.1007/978-3-319-22668-2_19
29. Magga, S.M.: Gákti on the pulse of time: the double perspective of the traditional sámi dress 1. In: The Sámi World, pp. 39–52. Routledge (2022)
30. Magnani, M., Guttorm, A., Magnani, N.: Three-dimensional, community-based heritage management of indigenous museum collections: archaeological ethnography, revitalization and repatriation at the sámi museum Siida. J. Cult. Herit. **31**, 162–169 (2018)
31. Micoli, L.L., Caruso, G., Guidi, G.: Design of digital interaction for complex museum collections. Multimod. Technol. Interact. **4**(2), 31 (2020)
32. Otto, K., Wood, K.: Product design: techniques in reverse engineering and new product development (2001)
33. Paananen, S., Suoheimo, M., Häkkilä, J.: Charting experience categories for museum exhibitions. In: Ardito, C., et al. (eds.) INTERACT 2021. LNCS, vol. 12936, pp. 372–376. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-85607-6_40
34. Paananen, S., Suoheimo, M., Häkkilä, J.: Decolonizing design with technology in cultural heritage contexts-systematic literature review. In: Bruyns, G., Wei, H. (eds.) With Design: Reinventing Design Modes. IASDR 2021. Springer, Singapore (2022). https://doi.org/10.1007/978-981-19-4472-7_119
35. Pallud, J.: Impact of interactive technologies on stimulating learning experiences in a museum. Inf. Manag. **54**(4), 465–478 (2017)
36. Parsons, M., Fisher, K., Nalau, J.: Alternative approaches to co-design: insights from indigenous/academic research collaborations. Curr. Opin. Environ. Sustain. **20**, 99–105 (2016)
37. Persad, U., Langdon, P., Clarkson, J.: Characterising user capabilities to support inclusive design evaluation. Univ. Access Inf. Soc. **6**(2), 119–135 (2007)
38. Peters, D., Hansen, S., McMullan, J., Ardler, T., Mooney, J., Calvo, R.A.: "participation is not enough" towards indigenous-led co-design. In: Proceedings of the 30th Australian Conference on Computer-Human Interaction, pp. 97–101 (2018)
39. Raskin, J.: Intuitive equals familiar. Commun. ACM **37**(9), 17–19 (1994)

40. Rello, L., Pielot, M., Marcos, M.C.: Make it big! the effect of font size and line spacing on online readability. In: Proceedings of the 2016 CHI conference on Human Factors in Computing Systems, pp. 3637–3648 (2016)
41. Saldaña, J.: The coding manual for qualitative researchers. SAGE, Los Angeles; London, 3rd edn. (2016). oCLC: ocn930445694
42. Schrepp, M., Hinderks, A., Thomaschewski, J.: Design and evaluation of a short version of the user experience questionnaire (ueq-s). Int. J. Interact. Multimed. Artif. Intell. **4**(6), 103–108 (2017)
43. Silvén, E.: Nomadising sami collections. In: Current Issues in European Cultural Studies; 15–17 June; Norrköping; Sweden 2011, pp. 271–275. no. 062, Linköping University Electronic Press (2011)
44. Silvén, E.: Constructing a sami cultural heritage. Ethnologia Scandinavica 44 (2014)
45. Tlostanova, M.: On decolonizing design. Design Philos. Papers **15**(1), 51–61 (2017)
46. Valkonen, S., Alakorva, S., Aikio, Á., Magga, S.M.: Introduction: introduction to the sámi world. In: The Sámi World, pp. 1–18. Routledge (2022)
47. Wang, G.G.: Definition and review of virtual prototyping. J. Comput. Inf. Sci. Eng. **2**(3), 232–236 (2002)
48. Webb, S.: Making museums, making people: the representation of the sámi through material culture. Public Archaeol. **5**(3), 167–183 (2006)
49. Winschiers-Theophilus, H.: Bridging worlds: indigenous knowledge in the digital world. In: Proceedings of the 22nd ACM/IEEE Joint Conference on Digital Libraries, pp. 1–2 (2022)
50. Wolf, K., Reinhardt, J., Funk, M.: Virtual exhibitions: what do we win and what do we lose? Electronic Visualisation and the Arts, pp. 79–86 (2018)
51. Xu, X., Kang, J., Yasin, I., et al.: Effect of music tempo on duration of stay in exhibition spaces. Appl. Acoust. **207**, 109353 (2023)
52. Yalamu, P., Chua, C., Doube, W.: Does indigenous culture affect one's view of an LMS interface: a PNG and pacific islands students' perspective. In: Proceedings of the 31st Australian Conference on Human-Computer-Interaction, pp. 302–306 (2019)
53. Zaman, T., Falak, H.: Framing indigenous knowledge in digital context: technologies, methods and tools. Int. J. End-User Comput. Develop. (IJEUCD) **7**(2), 36–51 (2018)